OPEN AGENT HACKATHON 2026 · TRACK 02: THE AGENT THAT CAN EXPLAIN WHY

WhyTrail

An agent that explains why a metric changed, and shows the query behind every number.

Team: Wei Fang Solo (one builder)

Sponsor tech: Zetaris Lightning SQL (every fact) · Meterless World Model + H-MEM (all state) · Claude (judgement) · built in Cursor

github.com/run58669-maker/whytrail · Apache-2.0 · `python -m whytrail --mock` runs offline in seconds

PROBLEM & TARGET USERS

"Why did the number change?" is the question nobody can answer with evidence

- Operations analyst, Monday morning: **"Manhattan taxi trips dropped last week. Why?"**
- The answer lives in three places that do not share a schema: **trip data, weather, calendar**.
- Today's options: a dashboard that shows *that* it dropped, or an LLM that *guesses* why.
- Neither shows the query behind the number. Neither can be audited.

Target user

Operations / BI analysts who must answer "why" with numbers someone else can check: SQL, source tables, row counts.

What they need

Competing explanations tested against the data, the losers shown as well as the winner, and every figure traceable to a query.

SOLUTION

Falsification-first: try to kill every explanation before you believe one

WhyTrail does not ask the model "why?". It asks the data **"which explanations survive?"**

- **1. List competing hypotheses**: holiday, weather, price, data gap, local-vs-citywide.
- **2. Run a query designed to refute** each one (Zetaris, federated).
- **3. A Skeptic attacks the survivors**: it names a follow-up test from a fixed catalog and the orchestrator runs exactly that test.
- **4. A Verifier re-derives the two headline totals** through an independent Zetaris query; a mismatch blocks the answer.
- **5. Only then is the answer written**, and every figure links to a fact: SQL, source tables, row count, result hash.

Output = answer + why-trail

The prose answer, the list of hypotheses with **supported** / **refuted** verdicts, the Skeptic's follow-up result, the Verifier lines, and an evidence ID per claim.

`f_b668b44f81` → SQL + tables + rows + hash in `logs/world-<run>.jsonl`

AGENT WORKFLOW / ARCHITECTURE

Six agents in a bounded loop, not a fixed chain

```
question → Orchestrator (bounded state machine, JSONL trace)
  H-MEM hydrate (Meterless) → Planner: hypotheses, each with a falsifying test
  Retriever: SQL templates → Zetaris REST → World Model (provenance per fact)
  Analyst: deterministic verdict per hypothesis (supported / refuted)
  Skeptic: strongest alternative + follow-up test from FOLLOWUP_TESTS
       isolate_daytype    : federated join NY_TAXI x NOAA_GHCN x REF, weekday-matched
       baseline_variation : total of the period before A; ordinary variation?
       → back to Retriever; a failed hypothesis is downgraded: needs_revision
  Verifier: second Zetaris query re-derives the two headline totals (A, B);
       a mismatch blocks the Writer and replaces the number
  Writer: answer + why-trail from World Model payloads only
  H-MEM mine: verdicts persisted for the next run
```

- **Planner, Skeptic, Writer** call the model through **one injected generator**.
- **Retriever, Analyst, Verifier** are plain code: numbers never come from model text.
- Real back-edges: Skeptic → Retriever; Verifier can block the Writer; H-MEM changes the next Planner run.
- Every step is a line in `logs/trace-<run_id>.jsonl`: agent, action, fact ID, elapsed time.

Zetaris for every fact, Claude for judgement, Meterless engines for state

Zetaris Lightning SQL

data harness

- Direct REST API `POST /api/v1.0/sql-editor/sqls/run`: bearer key, `X-Org-ID`, UUID request IDs, one statement per call.
- Three sources with qualified names: `NY_TAXI.taxi_daily`, `NOAA_GHCN.central_park_daily`, `REF.us_holidays`; the Skeptic's `isolate_daytype` test is a federated join across all three.
- Dataset reproducible from `scripts/zetaris/` (export, clean, register).

Claude

- `claude-opus-5-5` via the Anthropic SDK behind one injected generator; deterministic mock generator for offline sample mode.

Meterless

agent engines, in-repo

- **World Model**: WorldStore, IngestPipeline (idempotent, provenance), ContextBuilder. Evidence ledger: written by Retriever/Analyst/Verifier, read by Skeptic/Writer.
- **H-MEM**: MemoryStore (layers, supersede, persist), MemoryRetrieval (hybrid score, retrievalReason). Schema facts + verdicts across runs; read by the Planner.
- Engines never call a model.

Runtime & tooling

- Python 3.11+, CLI `python -m whytrail` and HTTP `POST /run`, Dockerfile, **21 unit tests**, `SAMPLE_MODE=true` fallback.
- Built in **Cursor** (`.cursor/rules/whytrail.mdc`; Cursor commits tagged `[cursor]`).

MLK week, −1.42%: the holiday explains 100%, weather is refuted

```
$ python -m whytrail --memory .whytrail/hmem-live.json --reset-memory --max-cost 0.5
[INFO] run-b5402d4e Orchestrator.parse: segment=Manhattan A=2025-01-13.. B=2025-01-20..
[INFO] run-b5402d4e Analyst.verdict: supported: holidays in period: 0 -> 1
[INFO] run-b5402d4e Analyst.verdict: refuted: snow 41mm -> 0mm, precipitation 9mm -> 0mm
[INFO] run-b5402d4e Skeptic.request_followup: 1 challenges
[INFO] run-b5402d4e Verifier.check: metric_a: cited 758,672 vs re-derived 758,672 -> ok
[INFO] run-b5402d4e Verifier.check: metric_b: cited 747,921 vs re-derived 747,921 -> ok

=== WhyTrail answer ===
Manhattan taxi trips fell 1.42%, from 758,672 in the week of 2025-01-13 to 747,921
in the week of 2025-01-20. The drop is explained by Martin Luther King Jr. Day on
2025-01-20, which had only 71,364 trips compared with 88,147 the Monday before, a
loss of 16,783 trips. The other six days of the holiday week actually gained 6,032
trips combined, so the holiday accounts for 100% of the decline (confidence 0.95).

=== why-trail ===
- H_holiday  -> supported: holidays 0 -> 1; follow-up: holiday days explain 100%
                           [evidence f_b668b44f81, f_75f1825a48]
- H_weather  -> refuted: snow 41mm -> 0mm         [evidence f_60fc69bd8d, f_4858832554]
- H_data_gap -> refuted: days with data 7 -> 7    [evidence f_9457ecdef0, f_34ab0fd871]
- H_price    -> refuted: fare $13.87 -> $13.72    [evidence f_a9d5a89340, f_e4dd8bb6be]
- H_local    -> refuted: Manhattan -1.4% vs others +8.5%  [f_ef7be50d2f, f_52495cef71]
- verify:metric_a: 758,672 | re-derived 758,672 from raw rows -> ok  [f_0203d716c5]
- verify:metric_b: 747,921 | re-derived 747,921 from raw rows -> ok  [f_cf3eb3b025]

zetaris_queries=20 model_calls=3 cost=$0.0565 world_facts=20 hmem_records=7 seconds=34.28
$ python -m whytrail --quiet "Why did Manhattan taxi trips drop in the week of 2024-01-15 ..."
ERROR NoDataError: no NY_TAXI.taxi_daily rows for Manhattan in 2024-01-08..2024-01-14 and
2024-01-15..2024-01-21; available data: 2024-04-01..2025-02-28. Ask about weeks inside that range.
```

20

Zetaris queries, incl. 2 Verifier re-checks

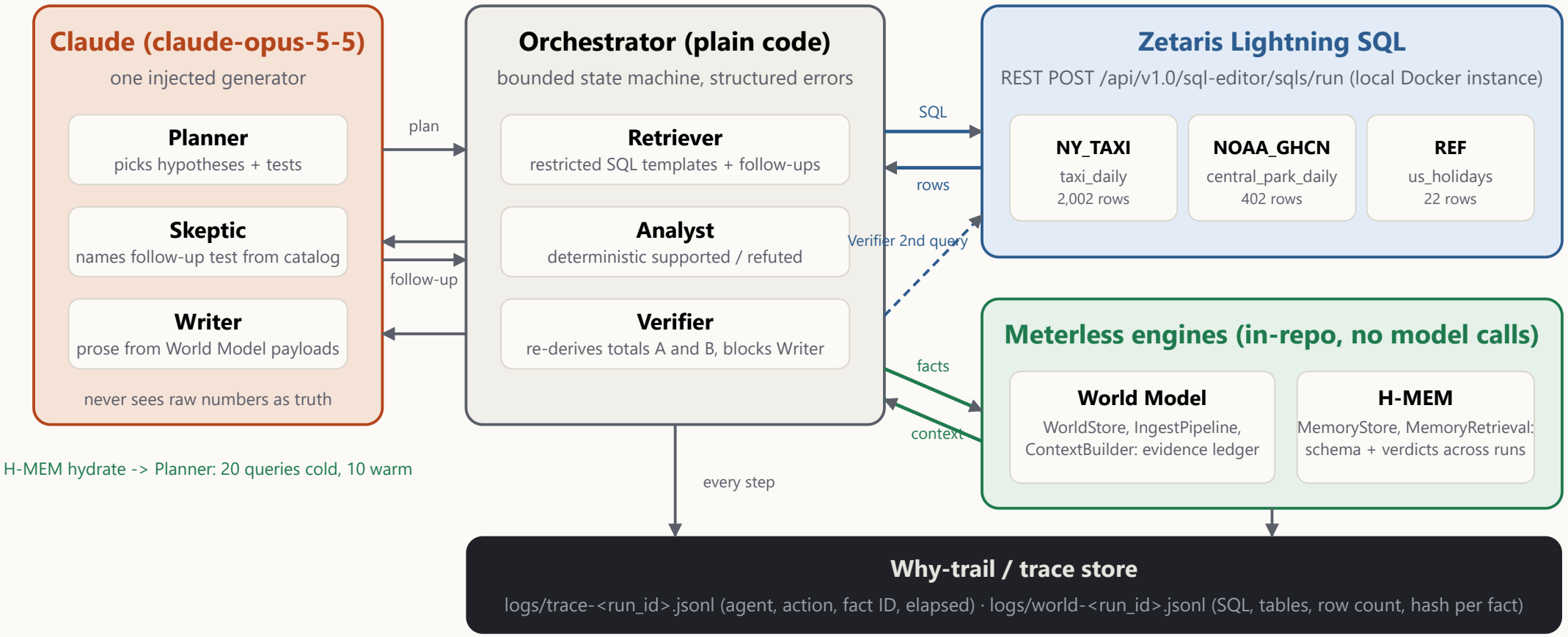
3

Claude calls: Planner, Skeptic, Writer

34.3 s · $0.0565

wall time and model cost for the run

Claude agents ↔ Zetaris Lightning SQL ↔ Meterless engines, with the why-trail store



The model proposes what could be false; Zetaris decides

- **Evidence is a first-class object.** Every number in the answer resolves to a World Model fact with SQL, tables, row count and hash. No fact, no sentence.
- **Zetaris closes its own loop.** The two headline totals are queried twice with two formulations (template aggregate vs raw rows summed client-side); the Verifier is an independent reader of the same federated data.
- **The Skeptic's follow-up is itself a federated query**: NY_TAXI × NOAA_GHCN × REF, weekday-matched, inside Zetaris, not stitched in Python. The Skeptic names the test; the orchestrator runs exactly that and traces it.
- **H-MEM measurably changes behaviour**: same question twice in sample mode, **20 queries cold, 10 warm** (schema discovery skipped, refuted hypotheses cited from memory). `examples/run1_cold_memory.txt`, `run2_warm_memory.txt`.
- **Numbers never come from model text.**

Honest limits (kept consistent with the repo)

- Fixed catalogs: 5 hypotheses, 2 follow-up tests (`isolate_daytype` , `baseline_variation`). The model chooses, it does not invent.
- SQL restricted to templates on three tables; question parser fixed to taxi trips, a borough, two ISO-dated weeks.
- Verifier re-derives the **two headline totals only**; other numbers carry provenance, not a re-derivation.
- Sample mode = synthetic Jan 2024 data, not real TLC/NOAA. Live range 2024-04-01..2025-02-28; weather through 2025-02-05.
- One recorded live run (run-b5402d4e), made before the follow-up catalog existed: its Skeptic asked for `isolate_daytype` , the only test then available.
- NVIDIA not used yet. Cursor = dev environment, not runtime.

What's next, and where to find everything

Next

- Web UI for the why-trail (click a number, see its SQL and rows).
- Route the Skeptic to **NVIDIA Nemotron** through the same injected generator.
- Open the hypothesis and follow-up catalogs: model-proposed tests validated against the template grammar.
- Extend the Verifier beyond the two headline totals.
- Docker image test from a clean clone; more domains on the same three-source pattern.

Links

Repo: **github.com/run58669-maker/whytrail** (Apache-2.0)

Run offline, no keys: `python -m whytrail --mock` (after `pip install -r requirements.txt`)

Tests: `python -m pytest` (21 tests, offline, ~2 s)

Live transcript: `examples/run5_live_zetaris_claude.txt`
Trace: `logs/trace-run-b5402d4e.jsonl`

Dataset reproduction: `scripts/zetaris/` · Deck source: `docs/deck/index.html`

Team: Wei Fang Solo · Track 02 · Pre-existing components: none (all code written after 2026-10-22 00:00 UTC)